

On the number of Sudoku squares

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ARTICLE INFO

Article history:

Received 20 April 2018
Received in revised form 25 July 2018
Accepted 6 August 2018
Available online 5 September 2018

Keywords:

Latin squares
Sudoku squares
Permanents
Upper bounds on permanents

ABSTRACT

We provide an upper bound on the number of $n^2 \times n^2$ Sudoku squares, and explain intuitively why there is reason to believe that the bound is tight up to a multiplicative factor of a much smaller order of magnitude. A similar bound is established for Sudoku squares with rectangular regions.

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1. Introduction

The Sudoku puzzle is a very popular puzzle, if not the most popular one, in recent years. It (usually) consists of a 9×9 grid, some of whose cells are marked by numbers between 1 and 9. One is required to fill in all the empty cells, again by numbers between 1 and 9, with the following constraints:

- Each row contains each of the numbers $1, 2, \dots, 9$ (each appearing exactly once).
- Each column contains each of the numbers $1, 2, \dots, 9$.
- Each 3×3 mini-grid (see Fig. 1) contains each of the numbers $1, 2, \dots, 9$.

Thus, a (filled) Sudoku square is a special Latin square of order 9. A *Latin square* of order n is an $n \times n$ grid, whose cells are marked by numbers between 1 and n , satisfying the analogues of the first two properties above. Namely, each row and column contains each number exactly once.

Latin squares have been studied extensively from various viewpoints; see, for example, [30]. In particular, there is a lot of literature regarding the number L_n of Latin squares of order n . Euler [5] found this number for $n = 5$ in 1782. Due to the very fast growth of the sequence, our knowledge of L_n for larger sizes of n has not progressed much in the centuries that elapsed. In fact, the largest n for which the value of L_n is currently known exactly is $n = 11$; see [19]. For all values up to 11, see Sequence A002860 in the On-Line Encyclopedia of Integer Sequences [27].

The situation is much better when it comes to estimates of L_n . The following bounds were obtained for L_n (cf. [30, Ch. 17]):

$$n!^{2n}/n^{n^2} \leq L_n \leq \prod_{k=1}^n k!^{n/k}. \quad (1)$$

Note that both bounds have a roughly similar order of magnitude; in fact, employing Stirling's formula

$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \cdot (1 + o(1)),$$

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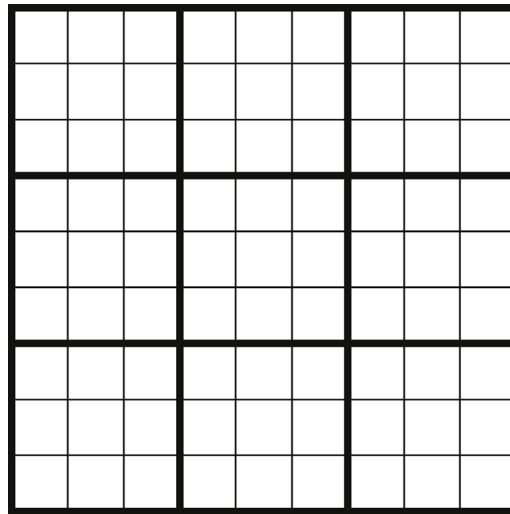


Fig. 1. An empty Sudoku grid.

and routine estimates, we get an effective constant C such that

$$\left(\frac{n}{e^2}\right)^{n^2} e^{n \log n} \leq L_n \leq \left(\frac{n}{e^2}\right)^{n^2} e^{C n \log^2 n}. \quad (2)$$

For more information on many aspects of Latin squares (and rectangles) we refer the reader to the survey paper [29].

The mathematics of Sudoku has also been considered from several angles. We refer to [12] for general information on the puzzle, and to [8,25] for discussion of relevant questions of a mathematical nature, and in particular enumerative problems. The question as to the minimum number of clues that must be initially provided, so that the puzzle has a unique solution, has attracted a lot of attention. Many thousands of uniquely solvable 17-clue puzzles have been found [23,24], but only recently has it been verified that there exists no such 16-clue puzzle [18].

As for Latin squares, Sudoku squares admit an immediate generalization to other sizes. An $n^2 \times n^2$ Sudoku grid consists of a grid of this size, in which each row, each column, and each of the $n \times n$ mini-grids composing the whole grid, includes all numbers between 1 and n^2 . Our main interest here is in the number S_n of such squares. A further generalization of Sudoku is rectangular Sudoku, in which the mini-grids are not necessarily squares but any rectangles. (However, we emphasize that the whole board is still a square.) Let R, C be positive integers. An $R \times C$ rectangular Sudoku is an $RC \times RC$ square, whose cells are marked by numbers between 1 and RC , satisfying the row and column constraints, and divided into RC rectangles of size $R \times C$, in each of which again all the numbers appear exactly once. The case $R = C = n$ is that of classical Sudoku. The case where either $R = 1$ or $C = 1$ reduces to that of a Latin square. We will also be interested in the number $S_{R,C}$ of $R \times C$ rectangular Sudoku squares.

Similarly to Latin squares, the rate of growth of S_n makes it very hard to calculate it exactly. While $S_1 = 1$ and $S_2 = 288$, the next term of the sequence, S_3 , is approximately $6.67 \cdot 10^{21}$ (see [7,28]), and is the last term known exactly. In fact, except for S_2 , which is trivial to calculate, and $S_{2,3}$, which has been calculated “mathematically” [14], the other known values of $S_{R,C}$ have been calculated employing heavy computations. (For more information we refer to [15].) There are also some heuristics that may be used to estimate S_n , and more generally $S_{R,C}$, that give good approximations in the cases where the exact value is known; see [13] and the references there. There has been even an attempt to estimate S_n by stochastic simulation [22]. Unfortunately, this is a rare-event simulation; unless n is very small, it cannot provide good estimates in reasonable time.

Herzberg and Murty [11] dealt with the question, and were able to bound S_n from above as follows:

$$S_n \leq n^{2n^4} e^{-2.5n^4 + O(n^3 \log n)}. \quad (3)$$

Clearly, $S_n \leq L_{n^2}$. By the right-hand side inequality in (2), this gives:

$$S_n \leq n^{2n^4} e^{-2n^4 + O(n^2 \log^2 n)}. \quad (4)$$

Thus, Herzberg and Murty’s bound of (3) improves the trivial bound of (4) by a factor of roughly $e^{n^4/2}$. (We refer to [26] for a careful derivation of the improved bound.)

Our main result in this paper, stated in Section 2, is an improvement of (3). We also present a lower bound on S_n , that is probably much further from the true value than is our upper bound. A version of the upper bound for rectangular Sudoku is stated also. In Section 3 we discuss some upper bounds for permanents of matrices. These bounds are the main technical

tool in the proof of the main result. The proof of the main result itself is presented in Section 4. In Section 5 we derive our lower bound. Finally, in Section 6, we will argue on intuitive grounds that this bound is in fact “roughly tight”. Based on this discussion, we will present a conjectured lower bound that is much closer to matching our upper bound, and a similar conjectured lower bound for the number of Latin cubes and hypercubes, studied by Linial and Luria [17].

We express our gratitude to M. Fried, M. Goldberg, A.F. Jarvis, S.K. Jones, R. Sharon and A. Zenou for their helpful comments in regard to this topic, and to the two referees of the first version on their numerous suggestions for improvement.

2. The main results

Our main result here is

Theorem 2.1. *The number of $n^2 \times n^2$ Sudoku squares is bounded above as follows:*

$$\begin{aligned} S_n &\leq \prod_{h=1}^n \prod_{k=1}^n (hk)!^{n^2/(hk)} \\ &= n^{2n^4} e^{-3n^4 + O(n^3 \log n)}. \end{aligned} \quad (5)$$

Namely, we obtain a reduction of the bound of Herzberg and Murty, given in (3), by a further factor of roughly $e^{n^4/2}$. In the other direction we get only

Proposition 2.2. *There exists an explicit constant $\lambda > 1$ such that*

$$S_n \geq \lambda^{n^4}, \quad n = 2, 4, 6, \dots$$

We now formulate the analogue of Theorem 2.1 for the case of Sudoku squares with rectangular regions.

Theorem 2.3. *The number of $R \times C$ rectangular Sudoku squares (where $R, C \geq 2$) is bounded above as follows:*

$$\begin{aligned} S_{R,C} &\leq \prod_{h=1}^C \prod_{k=1}^R (hk)!^{RC/(hk)} \\ &= (RC)^{(RC)^2} e^{-3(RC)^2 + O(M^2 m \log m)}, \end{aligned}$$

where $M = \max\{R, C\}$, $m = \min\{R, C\}$.

The proof of Theorem 2.3 is the same as that of Theorem 2.1, and we omit it.

3. Upper bounds on permanents

In this section we discuss several upper bounds on the permanent of a matrix, given some partial information about the matrix. These bounds will serve as the main tool in the proof of Theorem 2.1.

Recall that the permanent $\text{per}(M)$ of a matrix $M = (m_{ij})_{i,j=1}^n$ is defined by

$$\text{per}(M) = \sum_{\sigma} \prod_{i=1}^n m_{i,\sigma_i},$$

where the product extends over all permutations σ of $\{1, 2, \dots, n\}$. For more information on permanents we refer to [20]. The main tool we will use is the following result, due to Brègman [3].

Brègman's Theorem. *Let $M = (m_{ij})_{i,j=1}^n$ be a $(0, 1)$ -matrix, such that the number of 1-entries in each row i is r_i . Then*

$$\text{per}(M) \leq \prod_{i=1}^n r_i!^{1/r_i}.$$

We note in passing that there are several other bounds on the permanent, that are sometimes tighter [2,16,32], but they do not seem to yield better results in our situation.

We are going to prove a version of Brègman's theorem. Here we deal with $(0, 1)$ -matrices in which the number of 1s in each row is not given. Rather, only the total number of 1s in the whole matrix is known. Roughly speaking, we claim that, out of all possibilities of splitting the prescribed number of 1s between the rows of the matrix, the worst bound given by Brègman's theorem is in the case where there is the same number of 1s in all rows. More formally (and including the case where the average number of 1s per row is not integer), we prove

Proposition 3.1. Let $M = (m_{ij})_{i,j=1}^n$ be a $(0, 1)$ -matrix, such that the total number of 1s in the whole matrix is R . Then

$$\text{per}(M) \leq \lfloor R/n \rfloor!^{\frac{n-R \bmod n}{\lfloor R/n \rfloor}} \lceil R/n \rceil!^{\frac{R \bmod n}{\lceil R/n \rceil}}. \quad (6)$$

Here, $\lfloor x \rfloor$ (resp., $\lceil x \rceil$) denotes the largest (resp., smallest) integer not larger (resp., smaller) than a real number x , and $a \bmod b$ denotes the least non-negative residue of a modulo b .

If the number R in the proposition is a multiple of the order of the matrix (which will turn out to be the case in our application of the proposition), the proposition assumes the following, very simple form.

Corollary 3.2. Let $M = (m_{ij})_{i,j=1}^n$ be a $(0, 1)$ -matrix, such that the total number of 1s in the whole matrix is rn for some integer r . Then

$$\text{per}(M) \leq r!^{n/r}.$$

Remark 3.3. 1. It may be the case that the right-hand side of (6) can be replaced in general (i.e., for r not necessarily an integer) by the right-hand side of the last inequality (with $\Gamma(r+1)$ instead of $r!$), but we have not checked it.

2. The maximum permanent itself, under the given constraint, may be obtained (at least also) for matrices with imbalanced numbers of 1s at the various rows. For example, Corollary 3.2 ensures that the permanent of a 3×3 matrix, with six 1s and three 0-s, is at most $2!^{3/2} = 2\sqrt{2}$, so that it is at most 2. The two matrices

$$M_1 = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

both have six 1s, and $\text{per}(M_1) = \text{per}(M_2) = 2$. Yet, M_1 has the same number of 1s in each row, while M_2 does not.

The main ingredient in the proof of Proposition 3.1 is given by the following lemma, due to Wanless [31]. (See the proof of Lemma 1 there; see also [1, Thm. 1] for a proof not involving calculus.)

Lemma 3.4 ([31]). The function $g : \{1, 2, \dots\} \rightarrow \mathbf{R}$, given by

$$g(n) = (n+1)!^{1/(n+1)} / n!^{1/n}$$

for all n , is strictly decreasing.

Proof of Proposition 3.1. Let r_i be the number of 1s in each row i of M . Without loss of generality, $r_1 \geq r_2 \geq \dots \geq r_n$. If $r_1 \leq r_n + 1$, then the bound to prove coincides with the bound given by Brègman's theorem. If $r_1 \geq r_n + 2$, then by Lemma 3.4

$$r_1!^{1/r_1} r_2!^{1/r_2} \dots r_n!^{1/r_n} < (r_1 - 1)!^{1/(r_1-1)} r_2!^{1/r_2} \dots (r_n + 1)!^{1/(r_n+1)}.$$

Employing Lemma 3.4 successively until all factorials are of numbers differing by at most 1 from each other, we arrive at the required conclusion. ■

4. Proof of Theorem 2.1

The naive approach for finding the number of Latin squares, or Sudoku squares, is to fill the square row after row, and count the possibilities for filling in each row. However, one readily realizes that the number of possible fillings of a row depends non-trivially on the way former rows have been filled. Thus, one tries instead to find upper and lower bounds on the number of fillings of each row, given that of the former rows. The implementation, in the case of Latin squares, is as follows. Suppose we have filled in the first k rows, for some $0 \leq k \leq n-1$, in a legal manner (i.e., no equal cells in any row and in any column). We need to fill in the $(k+1)$ st row. To this end, we define an $n \times n$ matrix $A = (a_{ij})_{i,j=1}^n$ of 0-s and 1s, that encodes the possibilities. Each entry a_{ij} of A is 1 if it is allowed to place the number j at the i th cell of row $k+1$ of the Latin square we fill (that is, j has not been used at the i th cell of any of the rows $1, 2, \dots, k$), and is 0 otherwise. It is easy to verify that the number of possible fillings of row $k+1$ of the Latin square is exactly the permanent $\text{per}(A)$ of A . Thus, good estimates on the permanents of the matrices we get in the process, at all steps $k \in \{0, 1, \dots, n-1\}$, would yield good estimates on L_n .

This approach indeed works very well. It is easy to see that, in the matrix A that encodes our possibilities in the $(k+1)$ st row of the Latin square, the number of 1s is $n-k$ both in every row and in every column. Thus, A is a scalar multiple of a bistochastic matrix, so that the lower bound established by Falikman [6] and by Egorychev [4] in their proofs of van der Waerden's conjecture yields a lower bound on $\text{per}(A)$. In the other direction, Brègman's theorem readily yields an upper bound on $\text{per}(A)$. Jointly, these bounds produce the two bounds in (1) on L_n .

For Sudoku squares, this approach totally fails when it comes to a lower bound. For example, while the partially filled Sudoku square in Fig. 2 does not violate so far any of the requirements, one cannot fill in the first cell of row 6 legally. Namely, the fact that a number of rows of a Sudoku square have been filled legally, does not guarantee that there is even a single

1	2	3	4	5	6	7	8	9
4	7	9	2	8	1	3	5	6
8	6	5	9	7	3	4	1	2
5	3	2	1	4	9	8	6	7
6	9	7	8	2	5	1	3	4

Fig. 2. A legal partial Sudoku square without completion.

way to complete the square. We mention in passing that the same problem occurs when one tries to bound from below the number of Latin cubes (or hypercubes) of order n (see Linial and Luria [17]).

The approach does apply, though, to bound from above the number of possibilities of filling in any row of a Sudoku square. However, there is a complication. Namely, the situation exemplified by Fig. 2 shows that the number of options for filling a cell of a row of the Sudoku square is not the same for all cells of that row. For Latin squares, there were $n - k$ options for each cell of row $k + 1$, and thus had an upper bound of $(n - k)!^{n/(n-k)}$ on $\text{per}(A)$, which is the number of possibilities of filling the whole row. How do we bound $\text{per}(A)$ now, when we do not know the number of 1s in each row of A ?

Suppose we are now filling in the $(k + 1)$ st row of the $(h + 1)$ st band. (Here, a band is a contiguous block of n rows.) Each cell in this row must be distinct from both the kn cells already filled in the $n \times n$ mini-grid containing our cell, as well as from the hn cells already filled above this cell in the h preceding bands. (We disregard here the k cells immediately above our cell, as they have already been counted among the former kn .) Now, while the former kn constraints are mutually distinct, as are the latter hn , one should certainly expect in general to have non-trivial overlapping between the 2 sets. For example, referring again to the example depicted in Fig. 2, there are 9 distinct constraints for the first cell of row 6, but only 6 constraints for the second cell. Namely, the number of active constraints may be anywhere between $\max\{kn, k + hn\}$ and $(k + h)n$. How can we use this information optimally?

Herzberg and Murty [11] used the lower bound $\max\{kn, k + hn\}$ on the number of mutually distinct constraints at every cell. This estimate yields the bound in (3) on S_n . Here we propose two alternatives, that yield the better bound of (5).

We first observe that, while the number of 1s at each row of the matrix A is not uniquely determined by k and h , the number of 1s at each column is in fact uniquely determined. Indeed, consider any number $s \in \{1, 2, \dots, n^2\}$. Clearly, the occurrences of s at the first k rows of the $(h + 1)$ st band of the square we fill in are all at distinct $n \times n$ mini-grids, and hence yield kn restrictions on occurrences of s in the next row. Now consider any of the other $n - k$ mini-grids. The number s appears once at each of the h mini-grids above it, and these occurrences are at distinct columns. Hence there are $(n - k)h$ additional locations along the row where s is not allowed. Altogether, there are

$$n^2 - (kn + (n - k)h) = (n - h)(n - k)$$

locations along the $(k + 1)$ st row of the $(h + 1)$ st band where s may appear. Thus, the number of 1s in each column of A is $(n - h)(n - k)$. Since Brègman's theorem clearly holds if rows are replaced by columns, we get that $\text{per}(A)$, which is the number of possible fillings of the current row of the Sudoku square, is at most

$$\prod_{i=1}^{n^2} ((n - h)(n - k))^{1/((n - h)(n - k))} = ((n - h)(n - k))^{n^2/((n - h)(n - k))}. \quad (7)$$

The other option we suggest is by using the rows of A . Since the number of 1s in each column of A is $(n - h)(n - k)$, the average number of 1s in the rows is also $(n - h)(n - k)$. Thus, we may use Brègman's theorem with row lengths $r_1, r_2, \dots, r_{n^2}$ that are not known, but with known sum $\sum_{i=1}^{n^2} r_i = (n - h)(n - k)n^2$. Corollary 3.2 gives again the upper bound in (7) on $\text{per}(A)$.

Now we can conclude the proof of [Theorem 2.1](#). The discussion above and (7) yield

$$S_n \leq \prod_{h=0}^{n-1} \prod_{k=0}^{n-1} ((n-h)(n-k))!^{n^2/((n-h)(n-k))}.$$

A change of variables produces a somewhat simpler expression:

$$S_n \leq \prod_{h=1}^n \prod_{k=1}^n (hk)!^{n^2/(hk)}.$$

Routine calculations, and use of Stirling's formula, give:

$$\begin{aligned} \log S_n &\leq n^2 \cdot \sum_{h=1}^n \sum_{k=1}^n \frac{1}{hk} \log(hk)! \\ &= n^2 \cdot \sum_{h=1}^n \sum_{k=1}^n \frac{1}{hk} [hk \log(hk) - hk + O(\log(hk))]. \end{aligned}$$

By the symmetry of h and k on the right-hand side:

$$\begin{aligned} \log S_n &\leq n^2 \cdot \sum_{h=1}^n \sum_{k=1}^n \left[2 \log k - 1 + \frac{1}{hk} O(\log k) \right] \\ &\leq 2n^3 \log n! - n^4 + n^2 \cdot \sum_{h=1}^n \frac{1}{h} \sum_{k=1}^n \frac{1}{k} \cdot O(\log n) \\ &= 2n^4 \log n - 2n^4 + O(n^3 \log n) - n^4 + n^2 \cdot O(\log^3 n) \\ &= 2n^4 \log n - 3n^4 + O(n^3 \log n). \end{aligned}$$

Exponentiating both sides, we obtain the required expression. ■

Remark 4.1. While the first approach in the proof was simpler, the second approach may have its advantages. First, it may be useful in other situations, where the number of 1s is the same neither in all rows nor in all columns of the matrix encoding the legal options. Moreover, it is natural to expect that, in the process of filling in the rows of the Sudoku square, there will usually be some variance in the number of 1s between the rows of the encoding matrix. Thus, if we could understand the typical distribution of the r_i 's, we could in principle improve our bound on S_n .

5. Proof of the lower bound

Let n be even, say $n = 2m$. Starting with any $m^2 \times m^2$ Sudoku square, we will construct numerous $n^2 \times n^2$ Sudoku squares. It will be convenient to let the symbols in the initial square be all pairs (a, b) with $1 \leq a, b \leq m$ rather than all numbers between 1 and m^2 . We start with constructing one specific $n^2 \times n^2$ Sudoku square. Notice that the new square needs to be 4 times as long as the initial one, contain 4 times as many distinct symbols, and contain altogether 16 times as many cells. To this end, we are going to expand each $m \times m$ mini-grid into 4 mini-grids, of size $2m \times 2m$ each. First, each $m \times m$ mini-grid is replaced by 4 copies of itself, arranged as a 2×2 square of identical mini-grids. Now, take the 4 occurrences of any original cell in these 4 new mini-grids. Expand each such cell into a 2×2 square. If the original cell was marked by (a, b) , mark the 4 new cells by (a, b) , $(a, b + m)$, $(a + m, b)$ and $(a + m, b + m)$. Moreover, we require that the 4×4 square made of all these new 16 cells forms a $2^2 \times 2^2$ Sudoku square. (Note that this square does not consist of 16 adjacent cells, but rather of 4 separated 2×2 squares.)

Clearly, the procedure above generates an $n^2 \times n^2$ Sudoku square. Moreover, this square is made of $n^4/16$ subsquares, each of them 4×4 . Since $S_2 = 288$, each of the 4×4 subsquares can be filled in 288 ways, so that altogether we have constructed $288^{n^4/16}$ distinct $n^2 \times n^2$ Sudoku squares. This proves [Proposition 2.2](#) with $\lambda = 288^{1/16}$. ■

One can improve the constant λ at the expense of restricting n . In fact, we get these $288^{n^4/16}$ distinct $n^2 \times n^2$ Sudoku squares for each of the S_m initial Sudoku squares of order m . If m in the construction above was assumed to be even, then we would have a better lower bound on S_n than above. In the same vein, if n is of the form 2^r , then we get an even better λ . However, all this probably falls quite short of the true value of S_n , as we explain in the next section.

6. How tight are the bounds?

In this section we explain why we believe that the bound given by [Theorem 2.1](#) is accurate up to an $e^{o(n^4)}$ factor. Recall first that a *row-Latin square* of order n is an $n \times n$ grid, whose cells are marked by numbers between 1 and n , such that each

number appears exactly once in each row (see [21]). Column-Latin squares are defined analogously. Thus, letting $\mathcal{L}_n$, $\mathcal{R}_n$, $\mathcal{C}_n$ denote the sets of all Latin, row-Latin and column-Latin squares of order n , respectively, we have $\mathcal{L}_n = \mathcal{R}_n \cap \mathcal{C}_n$. Obviously, $|\mathcal{R}_n| = |\mathcal{C}_n| = n!$.

Suppose we choose a random $n \times n$ matrix T , each cell having equal probabilities of assuming each of the values $1, 2, \dots, n$, and distinct cells being independent. Clearly:

$$P(T \in \mathcal{L}_n) = \frac{L_n}{n^{n^2}}, \quad P(T \in \mathcal{R}_n) = P(T \in \mathcal{C}_n) = \frac{n!}{n^{n^2}}.$$

If the events $\{T \in \mathcal{R}_n\}$ and $\{T \in \mathcal{C}_n\}$ were independent, we would get that L_n is equal to $n!^{2n}/n^{n^2}$.

Now, while as a first approximation one may think that the two events in question are independent, a short reflection shows that they are in fact dependent. For example, if the matrix T is known to belong to $\mathcal{R}_n$, then, in particular, exactly n of its cells are 1, exactly n of them are 2, and so forth. This hints that the conditional probability $P(T \in \mathcal{C}_n | T \in \mathcal{R}_n)$ is larger than $P(T \in \mathcal{C}_n)$. Thus, one is led to believe that the events are positively correlated. The left-hand inequality in (1) indeed confirms that they are, although of course the proof is not probabilistic.

As an evidence that the intuition above yields reasonable initial estimates, let us note that a simple calculation shows that it leads to the guess that the number of $k \times n$ Latin rectangles is roughly $n!^{k+n}/(n^{kn}(n-k)!^n)$. Indeed, Godsil and McKay [10] proved that, for $k = o(n^{6/7})$, this is the main term (namely, up to a multiplicative factor of a much smaller order of magnitude). They conjectured [9] the same to hold as long as $k = O(n^{1-\delta})$.

Using the same intuition for Sudoku squares, we are led to expect a lower bound on S_n , as follows. This time we select uniformly and randomly an $n^2 \times n^2$ matrix with cells between 1 and n^2 , and ask for the probability of the intersection of the three events whereby: (i) the constraint on rows is satisfied, (ii) the constraint on columns is satisfied, and (iii) the constraint on $n \times n$ mini-grids is satisfied. The same intuition we employed for Latin squares leads now to

Conjecture 6.1. For every $n \geq 1$:

$$S_n \geq n^{2!^{3n^2}}/n^{4n^4}.$$

A routine calculation shows that the lower bound in the conjecture is $n^{2n^4}e^{-3n^4}$, up to an $e^{O(n^2 \log n)}$ factor. This is the same as the upper bound given by Theorem 2.1, except that the (multiplicative) error term there was $e^{O(n^3 \log n)}$.

We mention that the probabilistic arguments above have some resemblance to Kilfoil's approach for estimating $S_{R,C}$ (see [13]). Of course, that approach gives much more precise results, but does not yield an easy-to-use formula.

Using the same kind of reasoning, we state a parallel conjecture regarding the number of Latin hypercubes. Here, a d -dimensional Latin hypercube of order n is a function f from $\{1, 2, \dots, n\}^d$ into $\{1, 2, \dots, n\}$, with the property that

$$f(x_1, x_2, \dots, x_{i-1}, a, x_{i+1}, \dots, x_d) \neq f(x_1, x_2, \dots, x_{i-1}, b, x_{i+1}, \dots, x_d)$$

for every $1 \leq i \leq d$ and $x_1, \dots, x_{i-1}, a, b, x_{i+1}, \dots, x_d \in \{1, 2, \dots, n\}$ with $a \neq b$. Linial and Luria [17] studied the number of d -dimensional Latin hypercubes, and obtained an upper bound analogous (in magnitude) to the one established for Latin squares. They conjectured that the upper bound is in fact quite tight. We raise

Conjecture 6.2. For every dimension d and $n \geq 1$, the number $L_{d,n}$ of d -dimensional Latin hypercubes of order n satisfies:

$$L_{d,n} \geq n!^{dn^{d-1}}/n^{(d-1)n^d}.$$

The bound is $\left((1 + o(1))\frac{n}{e^d}\right)^n$, as in [17, Conjecture 1.3].

Another interesting question is how $S_{R,C}$ changes as R and C change. More precisely, suppose to begin with that R and C are such that RC is a square, say $RC = n^2$. How do $S_{R,C}$ and S_n compare? The first term $(RC)^{(RC)^2}$ in the asymptotic upper bound for $S_{R,C}$ reduces to the corresponding term n^{2n^4} in the asymptotic upper bound for S_n . The same holds for the second terms; $e^{-3(RC)^2}$ reduces to e^{-3n^4} . However, when it comes to the "error", the two expressions begin to differ. If R and C are roughly the same, then the expressions are similar. However, as R becomes small (say, constant, with C growing), the $O(M^2 m \log m)$ in the exponent becomes $O(n^4)$. For $R = 1$, this must be the case; otherwise we would get an impossible upper bound on the number of Latin squares. For other pairs (R, C) , it raises

Question 6.1. Suppose $1 \leq R_1 < R_2 \leq C_2 < C_1$, where $R_1 C_1 = R_2 C_2$. Is it necessarily the case that $S_{R_1, C_1} > S_{R_2, C_2}$?

Unfortunately, there is very little existing data on values of $S_{R,C}$ to test the question. The only relevant case is of $S_{2,6}$ vs. $S_{3,4}$. According to [13], we have $S_{2,6} \approx 3.8 \cdot 10^{49}$ while $S_{3,4} \approx 8.1 \cdot 10^{46}$.

Acknowledgement

Research supported in part by the Milken Families Foundation Chair in Mathematics.

References

- [1] M. Aaghabali, S. Akbari, S. Friedland, K. Markström, Z. Tajfirooz, Upper bounds on the number of perfect matchings and directed 2-factors in graphs with given number of vertices and edges, *European J. Combin.* 45 (2015) 132–144.
- [2] A.H. Al-Kurdi, Some upper bounds for permanents of $(0, 1)$ -matrices, *J. Interdiscip. Math.* 10 (2) (2007) 169–175.
- [3] L.M. Brègman, Certain properties of nonnegative matrices and their permanents, *Dokl. Akad. Nauk SSSR* 211 (1973) 27–30 (in Russian).
- [4] G.P. Egorychev, Proof of the van der waerden conjecture for permanents, *Sibirsk. Mat. Zh.* 22 (6) (1981) 65–71, 225 (in Russian). English translation: *Siberian Math. J.* 22 (1981) 854–859.
- [5] L. Euler, Recherches sur une nouvelle espèce de quarrés magiques, *Verh. V. H. Zeeuwsch Genootsch. Der Wetensch. Vliissingen* 9 (1782) 85–239.
- [6] D.I. Falikman, Proof of the van der waerden conjecture on the permanent of a doubly stochastic matrix, *Mat. Zametki* 29 (6) (1981) 931–938, 957 (in Russian). English translation: *Math. Notes Acad. Sci. USSR* 29 (6) (1981) 475–479.
- [7] B. Felgenhauer, A.F. Jarvis, Enumerating possible Sudoku grids, <http://www.afjarvis.staff.shef.ac.uk/sudoku/sudoku.pdf>.
- [8] B. Felgenhauer, A.F. Jarvis, Mathematics of Sudoku I, *Math. Spectr.* 39 (2006) 15–22.
- [9] C.D. Godsil, B.D. McKay, Asymptotic enumeration of Latin rectangles, *Bull. Amer. Math. Soc. (N.S.)* 10 (1) (1984) 91–92.
- [10] C.D. Godsil, B.D. McKay, Asymptotic enumeration of Latin rectangles, *J. Combin. Theory Ser. B* 48 (1) (1990) 19–44.
- [11] A.M. Herzberg, M.R. Murty, Sudoku squares and chromatic polynomials, *Notices Amer. Math. Soc.* 54 (6) (2007) 708–717.
- [12] <https://en.wikipedia.org/wiki/Sudoku>.
- [13] https://en.wikipedia.org/wiki/Mathematics_of_Sudoku.
- [14] S.K. Jones, S. Perkins, P.A. Roach, On the number of 6×6 Sudoku grids, *J. Combin. Math. Combin. Comput.* 89 (2014) 33–43.
- [15] S.K. Jones, S. Perkins, P.A. Roach, On the number of sudoku grids, *Math. Today (Southend-on-Sea)* 50 (2) (2014) 94–95.
- [16] H. Liang, F. Bai, An upper bound for the permanent of $(0, 1)$ -matrices, *Linear Algebra Appl.* 377 (2004) 291–295.
- [17] N. Linial, Z. Luria, An upper bound on the number of high-dimensional permutations, *Combinatorica* 34 (4) (2014) 471–486.
- [18] G. McGuire, B. Tugemann, G. Civario, There is no 16-clue Sudoku: solving the Sudoku minimum number of clues problem via hitting set enumeration, *Exp. Math.* 23 (2) (2014) 190–217.
- [19] B.D. McKay, I.M. Wanless, On the number of Latin squares, *Ann. Combin.* 9 (2005) 335–344.
- [20] H. Minc, . Permanents, With a Foreword By Marvin Marcus, *Encyclopedia of Mathematics and Its Applications*, vol. 6, Addison-Wesley Publishing Co., Reading, Mass., 1978.
- [21] D.A. Norton, Groups of orthogonal row-Latin squares, *Pacific J. Math.* 2 (1952) 335–341.
- [22] A. Ridder, Counting the number of Sudoku's by importance sampling simulation, in: *Proceedings of the Recreational Mathematics Colloquium III*, Assoc. Ludus, Lisbon, 2013, pp. 85–92.
- [23] G. Royle, Minimum Sudoku, <http://staffhome.ecm.uwa.edu.au/00013890/sudokumin.php>.
- [24] G. Royle, <http://staffhome.ecm.uwa.edu.au/00013890/sudoku17>.
- [25] E. Russell, A.F. Jarvis, Mathematics of Sudoku II, *Math. Spectr.* 39 (2006) 54–58.
- [26] B. Schaab, Finding bounds for the number of Sudoku squares, http://www.siu.edu/aweyhau/teaching/seniorprojects/schaab_final_paper.pdf.
- [27] N.J.A. Sloane, <http://oeis.org/A002860>.
- [28] N.J.A. Sloane, <http://oeis.org/A107739>.
- [29] D.S. Stones, The many formulae for the number of Latin rectangles, *Electron. J. Combin.* 17 (1) (2010) 1.
- [30] J.H. van Lint, R.M. Wilson, *A Course in Combinatorics*, second ed., Cambridge University Press, New York, 2001.
- [31] I.M. Wanless, Maximising the permanent and complementary permanent of $(0, 1)$ -matrices with constant line sum, *Discrete Math.* 205 (1–3) (1999) 191–205.
- [32] A.G. Weyhaupt, A note on some upper bounds for permanents of $(0, 1)$ -matrices, *J. Interdiscip. Math.* 12 (1) (2009) 123–128.